Question	Answer	Marks	Partial Marks
6(a)(i)	106.01 to 106.02	**4**	**M2** for [cos[$\angle CBD$] =] $\frac{192^2+168^2-287.9^2}{2\times192\times168}$ oe or **M1** for the implicit form **A1** for –0.276 to – 0.275
6(a)(ii)	292.0 or 291.98 to 291.99	**1**	
6(a)(iii)	310.0 or 310.03 to 310.04	**5**	**M2** for [sin A =] $\frac{168\times\sin(90-38)}{205.8}$ or **M1** for $\frac{\sin A}{168}=\frac{\sin(90-38)}{205.8}$ **A1** for [A =] 40.0 or 40.03 to 40.04 **M1 dep** for 270 + *their* angle DAB oe
6(b)(i)	15 500 or 15 501 to 15 503. …	**2**	**M1** for 0.5 × 192 × 168 × sin(106) oe
6(b)(ii)	55 400	**2**	**FT** 3.575 × *their* **(b)(i)** oe rounded to nearest 100 **M1** for figs 35 75 × figs *their* **(b)(i)** or figs 554 or figs 5541 to figs 5543
7(a)	25 36 10 15 35 51	**2**	**B1** for 3, 4 or 5 correct
7(b)	n^2	**1**	
7(c)(i)	92	**1**	
7(c)(ii)	$\frac{1}{2}(n^2-n)$ oe	**2**	**M1** for $\frac{1}{2}(3n^2-n)-n^2$ oe or for final quadratic answer with $\frac{1}{2}n^2$ oe or $-\frac{1}{2}n^2$ oe but not both

Question	Answer	Marks	Partial Marks
7(d)	$a = \frac{1}{2}$, $b = \frac{1}{2}$	**5**	**B2** for 2 correct equations eg $a + b = 1$, $8a + 4b = 6$ or **B1** for 1 correct equation **B2** for one correct value or **M1** (dep on at least B1) for correctly eliminating one variable from two linear equations in a and b OR **B2** for $a = \frac{1}{2}$ or **B1** for 6a = 3 or for 3rd difference = 3 **B2** for $b = \frac{1}{2}$ or **M1** for substituting *their* a into a correct equation of first differences
8(a)	$ab(3a - b)$ final answer	**2**	**B1** for $a(3ab - b^2)$ or $b(3a^2 - ab)$ or $ab(3a - b)$ seen
8(b)	$x > 7.5$ final answer	**2**	**B1** for $12+3 < 5x - 3x$ oe
8(c)	$27x^6y^{12}$	**2**	**B1** for two of 27, x^6 and y^{12} correct
8(d)	0.5 or $\frac{1}{2}$	**3**	**M2** for $4 = 6x + 2x$ or better or **M1** for $2(2 - x) = 6x$ oe
8(e)	$2x^3 + 5x^2 - 23x + 10$ final answer	**3**	**B2** for correct expansion of three brackets unsimplified **B1** for correct expansion of two brackets with at least 3 terms correct
8(f)(i)	$200\left(1+\frac{r}{100}\right)^2 = 206.46$ oe	**M1**	
	$1 + \frac{2r}{100} + \frac{r^2}{100^2}$ oe	**M1**	
	$r^2 + 200r - 323 = 0$	**A1**	Correct solution reached with no errors or omissions seen If 0 scored, **SC1** for $200(n)^2 = 206.46$